BEU CONNECT

Bihar Engineering University, Patna

B.Tech 8th Semester Exam., 2026

Code: 100817 | ***Time: 03 Hours***

Semester: VIII | ***Subject: CLOUD COMPUTING*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): ***[2 x 7 = 14]***

(a) Cloud computing mainly provides
- (i) Localized computing
- (ii) Shared computing resources
- (iii) Offline processing
- (iv) Manual scheduling

(b) Which service model provides virtual machines and storage?
- (i) SaaS
- (ii) PaaS
- (iii) IaaS
- (iv) DaaS

(c) A company requires 20 virtual machines each with 4 GB RAM. Total RAM required is
- (i) 40 GB
- (ii) 60 GB
- (iii) 80 GB
- (iv) 100 GB

(d) Which hypervisor runs directly on hardware?
- (i) Type-1 Hypervisor
- (ii) Type-2 Hypervisor
- (iii) Container Engine
- (iv) Virtual Switch

(e) If 5 tasks require 10 ms each and are executed sequentially, total execution time is
- (i) 10 ms
- (ii) 20 ms
- (iii) 50 ms
- (iv) 100 ms

(f) Spot instances in cloud computing are mainly used for
- (i) High-cost reserved workloads
- (ii) Low-cost temporary workloads
- (iii) Permanent storage
- (iv) Security enhancement

(g) Which cloud model is owned by a single organization?
- (i) Public Cloud
- (ii) Community Cloud
- (iii) Private Cloud
- (iv) Hybrid Cloud

(h) Fog computing mainly reduces
- (i) Latency
- (ii) Encryption
- (iii) Storage
- (iv) Virtualization

(i) The major security issue in multi-tenancy is
- (i) Data isolation
- (ii) Compilation error
- (iii) CPU speed
- (iv) Routing delay

(j) Advance reservation in cloud computing means
- (i) Dynamic scheduling
- (ii) Reserving resources before use
- (iii) Deleting resources
- (iv) VM shutdown

Q.2 ***[7+7=14]***

(a) Explain cloud computing architecture and deployment models with neat diagrams.

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(b) Differentiate between SaaS, PaaS, and IaaS with suitable examples.

Q.3 ***[7+7=14]***

(a) Explain virtualization in cloud computing and virtualization issues.

(b) Discuss multi-tenancy issues and their challenges.

Q.4 ***[7+7=14]***

(a) Explain the Amazon EC2 and Amazon S3 services and their architectures.

(b) Discuss the implementation of private and hybrid clouds using open-source tools.

Q.5 ***[7+7=14]***

(a) Discuss the architecture and working of hypervisors in cloud computing.

(b) Discuss different load balancing algorithms used in cloud computing.

Q.6 ***[7+7=14]***

(a) Explain resource optimization algorithms in cloud computing.

(b) Explain Quality of Service parameters in cloud computing.

Q.7 ***[7+7=14]***

(a) Explain task migration and VM migration techniques.

(b) Discuss cloud security threats and vulnerabilities.

Q.8 ***[7+7=14]***

(a) Discuss application-level, data-level, and VM-level security in cloud computing.

(b) Explain Green Cloud and IoT with suitable examples.

Q.9 Write short note on any two of the following: ***[7 x 2 = 14]***

(a) Mobile Cloud Computing

(b) Infrastructure Security

(c) SLA Management

(d) Fog Computing

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